Dr. Anil Kumar

(Ph.D. from IIT Roorkee)

Postdoctoral Researcher

Center for Computational Sciences, University of Tsukuba, Tsukuba, Ibaraki, Japan

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Position Held

01.01.2026 – present — **Postdoctoral Researcher**, Center for Computational Sciences, University of Tsukuba, 1 Chome Tennodai, Tsukuba, Ibaraki, Japan (Webpage).
Project: "**KAKENHI–Grant-in-Aid for Scientific Research (B)**", MEXT, Japan
Project Adviser: Prof. Noritaka Shimizu
Member: **ERATO TOMOE Three-Nucleon Force Project** research funding program by Japan Science and Technology Agency (JST) in Japan (Webpage).

01.12.2025 – present — **Visiting Scientist**, TRIP Headquarters, RIKEN, Japan (RIKEN Quantum Computing Project) (Webpage).

01.08.2023 – 31.12.2025 — **Postdoctoral Researcher**, Center for Computational Sciences, University of Tsukuba, 1 Chome Tennodai, Tsukuba, Ibaraki, Japan (Webpage).
Project: "**Program for promoting research on the supercomputer Fugaku**", MEXT, Japan
Project Adviser: Prof. Noritaka Shimizu
Member: **ERATO TOMOE Three-Nucleon Force Project** research funding program by Japan Science and Technology Agency (JST) in Japan (Webpage).

06.01.2023 – 10.07.2023 — **Assistant Professor**, Department of Physics, School of Science, JECRC University Jaipur, Rajasthan, India.

10.12.2021 – 08.03.2022 — **Research Associate**, Department of Physics, Indian Institute of Technology Roorkee, India.

Academic Qualifications

2016 – 2021 — **Ph.D. in Theoretical Nuclear Physics, Department of Physics, Indian Institute of Technology Roorkee (IIT Roorkee), India**
Date of Ph.D. Defense: 01.12.2021.
Broad Area: Nuclear Structure Theory
Thesis title: *Shell Model Study of Allowed and Forbidden Beta Decays*
Ph.D. Supervisor: Prof. Praveen C. Srivastava (Professor, IIT Roorkee)

Academic Qualifications (continued)

2013 – 2015 **M.Sc. Physics, Department of Physics, University of Rajasthan, Jaipur, Rajasthan, India.**
Special Emphasis on: Nuclear Physics, Quantum Mechanics, Classical Mechanics, Electrodynamics, Mathematical Physics, Condensed Matter Physics, and High-Energy Physics.
(First Division)

2010 – 2013 **B.Sc. with Physics, Mathematics and Chemistry, Shri Radheshyam R Morarka Government PG College, Jhunjhunu, (Affiliated: University of Rajasthan, Jaipur, Rajasthan, India).**
(First Division)

2008 – 2010 **Sr. Sec. School (XII), Board of Secondary Education Rajasthan (BSER), Rajasthan, India.**
(First Division)

2006 – 2008 **Sec. School (X), Board of Secondary Education Rajasthan (BSER), Rajasthan, India.**
(First Division)

Research Interests

- Theoretical nuclear physics, nuclear many-body systems, large-scale shell-model calculations, Monte Carlo Shell Model (MCSM), No-Core Monte Carlo Shell Model (NC-MCSM), quasiparticle vacua shell model (QVSM), nuclear structure and shell evolution, collective motion and nuclear deformation.

- *Ab initio* nuclear structure calculations, including the Valence-Space In-Medium Similarity Renormalization Group (VS-IMSRG) approaches using interactions derived from Chiral Effective Field Theory (ChEFT).

- Nuclear Astrophysics, Gamow–Teller transitions, allowed and forbidden β-decays, electron-capture rates, two-neutrino and neutrinoless double-beta ($2\nu\beta\beta$ and $0\nu\beta\beta$) decay, and effective values of the axial-vector coupling constant g_A.

- *Artificial Intelligence (AI)* and *Quantum Computing (QC)* Applications in Nuclear Structure and Shell-Model Calculations

Research Publications

Publications/Submissions in Peer-reviewed Journals:

1 **Structure of ^{28}Mg and influence of the neutron pf shell**,
J. Williams, ... , **Anil Kumar**, *et al.*,
Phys. Rev. C **100**, 014322 (2019).

2 **Ab initio description of collectivity for sd shell nuclei**,
Archana Saxena, **Anil Kumar**, Vikas Kumar, Praveen C. Srivastava, and Toshio Suzuki,
Hyperfine Interact **240:37** (2019).

Research Publications (continued)

3 **Second-forbidden nonunique β^- decays of ^{24}Na and ^{36}Cl assessed by the nuclear shell model**,
Anil Kumar, Praveen C. Srivastava, Joel Kostensalo, and Jouni Suhonen,
Phys. Rev. C **101**, 064304 (2020).

4 **Shell model results for nuclear β^--decay properties of sd shell nuclei**,
Anil Kumar, Praveen C. Srivastava, and Toshio Suzuki,
Prog. Theo. Exp. Phys. **2020** 033D01 (2020).

5 **Gamow-Teller transition strengths for selected fp shell nuclei,**
Vikas Kumar, Praveen C. Srivastava, and **Anil Kumar**,
Acta Physica Polonica B **51**, 4 (2020).

6 **High-spin states of ^{218}Th**,
D. O'Donnell, ... , **Anil Kumar**, *et al.*,
J. Phys. G: Nucl. Part. Phys. **47**, 095103 (2020).

7 **Second-forbidden nonunique β^- decays of ^{59,60}Fe: possible candidates for g_A sensitive electron spectral-shape measurements**,
Anil Kumar, Praveen C. Srivastava, and Jouni Suhonen,
Eur. Phys. J. A **57** 225 (2021).

8 **Shell-model description for the first-forbidden β^- decay of ^{207}Hg into the one-proton-hole nucleus ^{207}Tl**,
Anil Kumar, and Praveen C. Srivastava,
Nuclear Physics A, **1014** 122255, (2021).

9 **Structure of ^{46,47}Ca from the β^--decay of ^{46,47}K in the framework of nuclear shell-model**,
Priyanka Choudhary, **Anil Kumar**, Praveen C. Srivastava, and Toshio Suzuki,
Phys. Rev. C **103**, 064325 (2021) .

10 **Shell-model study for GT-strengths corresponding to β decay of ^{60}Ge and ^{62}Ge**,
Vikas Kumar, **Anil Kumar**, and Praveen C. Srivastava,
Nuclear Physics A **1017**, 122344 (2022).

11 **Emergence of an island of extreme nuclear isomerism at high excitation near ^{208}Pb**,
S. G. Wahid, S. K. Tandel, S. Suman, Praveen C. Srivastava, **Anil Kumar**, et al.,
Phys. Lett. B **832**, 137262 (2022).

12 **Shell-model description for the properties of the forbidden β^- decay in the region "north-east" of ^{208}Pb**,
Shweta Sharma, Praveen C. Srivatava, **Anil Kumar**, and Toshio Suzuki,
Phys. Rev. C **106**, 024333 (2022).

13 **Forbidden beta decay properties of ^{135,137}Te using shell-model**,
Shweta Sharma, Praveen C. Srivastava, and **Anil Kumar**,
Nuclear Physics A **1031**, 122596, (2023).

Research Publications (continued)

14 **Large-scale shell model study of β^--decay properties of $N = 126, 125$ nuclei: Role of Gamow-Teller and first-forbidden transitions in the half-lives**,
Anil Kumar, Noritaka Shimizu, Yutaka Utsuno, Cenxi Yuan, and Praveen C. Srivastava,
Phys. Rev. C **109**, 064319 (2024).

15 **Shell-model study for allowed and forbidden β^- decay properties in the mass region "south" of ^{208}Pb**,
Shweta Sharma, Praveen C. Srivastava, **Anil Kumar**, Toshio Suzuki, Cenxi Yuan, and Noritaka Shimizu,
Phys. Rev. C **110**, 024320 (2024).

16 **In-beam γ-spectroscopy of ^{217}Ac**,
D. Sahoo, A. Y. Deo, ... , **Anil Kumar**, *et al.*,
Phys. Rev. C **111**, 014318 (2025).

17 **Probing the shape evolution and shell structures in neutron-rich N=50 nuclei**
Anil Kumar, Noritaka Shimizu, Takayuki Miyagi, Yusuke Tsunoda and Yutaka Utsuno,
Phys. Lett. B **874**, 140184, (2026).

18 **NN and 3N Tensor Force in the $N = 34$ Shell Evolution: An Ab Initio Perspective**
Anil Kumar, Takayuki Miyagi, and Noritaka Shimizu,
arXiv:2605.16822 *(Submitted for publication in Physics Letter B (May 2026))*.

Refereed Publications in Conference Proceedings:

1 ***Ab initio* results of β^--decay properties of Z = 8-15 nuclei**,
Anil Kumar and Praveen C. Srivastava,
Proceedings of the DAE Symp. on Nucl. Phys. 63, 84 (2018).

2 **Shell model results of Gamow-Teller transition strengths for ^{48}Ti(^{3}He,t)^{48}V reaction**,
Vikas Kumar, **Anil Kumar**, Archana Saxena, and Praveen C. Srivastava,
Proceedings of the DAE Symp. on Nucl. Phys. 63, 106 (2018).

3 **Doubly open-shell nuclei with ab initio approaches**,
Archana Saxena, **Anil Kumar**, Vikas Kumar, and Praveen C. Srivastava,
Proceedings of the DAE Symp. on Nucl. Phys. 63, 102 (2018).

4 **Role of central, spin-orbit and tensor part of SDPF-MU interaction in Mg isotopes**,
Vikas Kumar, Praveen C. Srivastava, and **Anil Kumar**,
Proceedings of the DAE Symp. on Nucl. Phys. 64, 116 (2019).

5 **Shell model study of allowed and forbidden beta decays**,
Anil Kumar,
Proceedings of the DAE Symp. on Nucl. Phys. 65, 908 (2021).

6 **Study of intermediate excited states above the I=10^+ isomer in the ^{136}Ba nucleus**,
Suresh Kumar et al., **Anil Kumar**, and Praveen C. Srivastava,
Proceedings of the DAE Symp. on Nucl. Phys. 65, 42 (2021).

Research Publications (continued)

7 **First forbidden beta-decay properties of ^{137}Te using shell-model**,
Shweta Sharma, Praveen C. Srivastava, **Anil Kumar**,
Proceedings of the DAE Symp. on Nucl. Phys. 66, 76 (2022).

8 **Shell model study of first-forbidden beta decay around ^{208}Pb**,
P. C. Srivastava, S. Sharma, **Anil Kumar** and P. Choudhary,
J. Phys.: Conf. Ser. **2586**, 012026 (2023).

9 **Exploring the Structure of N = 50 Nuclei through the Monte Carlo Shell Model**,
Anil Kumar and Noritaka Shimizu,
Proceedings of the DAE Symp. on Nucl. Phys. **69**, 79 (2025).

Workshops/Conferences/Symposium Attended

1 International Workshop on Applications of Antineutrino Physics, IWAAP-2017, BARC Mumbai, India, 30 November- 01 December, 2017.

2 DAE-BRNS Symposium on Nuclear Physics, Thapar University, Patiyala, India, 20-24 December, 2017.

3 School Cum First Collaboration Meeting on Computational Nuclear Structure and Reactions (CMNSR-2018), SINP Kolkata, India, 2-22 January, 2018.

4 Attended and delivered a seminar in the international Conference on Neutrino and Nuclear Physics 2020 (CNNP2020), iThemba LABS, Cape Town (South Africa), 24-28 February, 2020.

5 Machine Learning and Data Analysis for Nuclear Physics, a Nuclear TALENT Course at the ECT*, Trento, Italy, 22 June - 03 July, 2020 (Online).

6 Advances in many-body theories: from first principle methods to quantum computing and machine learning, Workshop at ECT*, Trento, Italy, 02-06 November, 2020 (Online).

7 DAE-BRNS Symposium on Nuclear Physics, DAE Convention Centre, Mumbai, India, 1-5 December, 2021 (Online).

8 Nuclear Lifetimes, Transitions and Moments (NLTM2022), VECC, Kolkata, India, 1-3 February, 2022 (Online).

9 Frontiers in Nuclear Structure Theory, KTH, Stockholm, Sweden, 23-25 May, 2022 (Online).

10 Advancing physics at next RIBF (ADRIB24), RIKEN, Wako campus, Saitama, Japan, 23-24 January, 2024.

11 Reimei Workshop “Intersection of Nuclear Structure and Direct Reaction”, JAEA Tokai, Mito, Ibaraki, Japan, February 28- March 1, 2024.

Workshops/Conferences/Symposium Attended (continued)

12 The workshop on frontier nuclear studies with gamma-ray spectrometer arrays (gamma24), Osaka University, Osaka, Japan, 26–28 March 2024.

13 [RIBF ULIC mini-WS] Structure of neutron-rich matter revealed by beta decay, RIKEN, Tokyo, Japan, July 29-30, 2024.

14 "SNP-CNS Summer School 2024", RIKEN, Wako campus, August 3-8, 2024.

15 International Conference "Nuclear Theory in the Supercomputing Era NTSE–2024", Pukyong National University, Busan, South Korea, December 1-7, 2024.

16 "Symposium of Basic Science Research Programs on the Fugaku supercomputer", UrbanNet Kanda Conference, Tokyo, January 8-10, 2025.

17 "Symposium of Single-particle and collective motions from nuclear many-body correlation (PCM2025) ", Aizu Univesity, Aizu-wakamatsu, Japan, March 3-7, 2025.

18 "SNP-CNS Summer School 2024, RIKEN", Wako campus, August 22-28, 2025.

19 "DAE-BRNS Symposium on Nuclear Physics", NIT Jalandhar, December7-12, 2025.

20 Workshop on "Microscopic Description of Nuclear Collective Phenomena via Large-Scale Density Functional Calculations", Hokkaido University, Sapporo, Japan, January 5 - 9, 2026.

21 "Symposium of Basic Science Research Programs on the Fugaku supercomputer", Tokyo, Japan, January 13-15, 2026.

22 International Conference "Nuclear Theory in the Supercomputing Era NTSE–2026", IIT Roorkee, India, March 16-20, 2026.

23 "Third RIKEN Quantum internal meeting", RIKEN, Wako, Japan, March 30, 2026.

24 "The 5th Conference on Advances in Radioactive Isotope Science (ARIS2026)", Fukushima, Japan, June 1-7, 2026.

Talks Delivered/ Poster Presentations

1 **Contributed talk** on "Double and neutrinoless Double beta decays", at the School Cum First Collaboration Meeting on Computational Nuclear Structure and Reactions (CMNSR-2018), SINP Kolkata, India, as a part of young scientist session.

2 **Contributed talk** on "Theoretical description of half-lives and electron spectra for higher order forbidden non-unique β decays", at the international Conference on Neutrino and Nuclear Physics 2020 (CNNP2020), iThemba LABS,Cape Town (South Africa) on February 23, 2020 .

3 **Invited talk** on "Shell-model description of allowed and forbidden β decays", at GSI Helmholtzzentrum für Schwerionenforschung GmbH, Darmstadt, Germany (Online) on March 9, 2021.

Talks Delivered/ Poster Presentations (continued)

4 **Contributed talk** on “Shell-model study of allowed and forbidden Beta decays”, at conference of DAE-BRNS Symposium on Nuclear Physics on December 1, 2021.

5 **Invited talk** on “Shell-model study of allowed and forbidden Beta decays”, at Lanzhou University, China (Online) on February 21, 2022.

6 **Contributed talk** on “Shell-model studies in nuclear beta decays”, in Symposium “Frontiers in Nuclear Structure Theory” at KTH, Stockholm, Sweden, May 25, 2022 (Online).

7 **Invited talk** on “Shell-model studies in nuclear beta decays”, in Reimei Workshop “Intersection of Nuclear Structure and Direct Reaction” at Tokai Cultural Center, Tokai, Mito, Ibaraki, Japan, February 28, 2024.

8 **Poster present** on “Shell model study of β^--decay properties of $N = 126, 125$ nuclei” in the workshop on frontier nuclear studies with gamma-ray spectrometer arrays (gamma24), Osaka University, Japan, 27 March 2024.

9 **Invited talk** in [RIBF ULIC mini-WS] Structure of neutron-rich matter revealed by beta decay, RIKEN, Tokyo, Japan, July 29-30, 2024.

11 **Poster present** on “Shell model study of β^--decay properties of $N = 126, 125$ nuclei” in the 22^{nd} international conference "Recent Progress in Many-Body Theories (RPMBT 22)" CNS-SNP Summer School 2024, Riken Wako Campus, Tokyo, Japan, August 3-8, 2024.

12 **Poster present** on “Shell model study of β^--decay properties of $N = 126, 125$ nuclei” in the 22^{nd} international conference "Recent Progress in Many-Body Theories (RPMBT 22)" Tsukuba University, Tsukuba, September 23-27, 2024.

13 **Invited talk** in Rising Researchers Seminar Series (R2S2), Washington University, USA, November 12, 2024 (On Zoom).

14 **Invited talk** in International Conference “Nuclear Theory in the Supercomputing Era NTSE–2024”, Busan, South Korea, 1-7 December 2024.

15 **Contributed talk** on "Shell model study of beta-decay in neutron-rich N = 126, 125 nuclei along the r-process path", Symposium of Basic Science Research Programs on the Fugaku supercomputer, Tokyo, January 8-10, 2025.

16 **Poster present** on “Neutron-rich nuclei around ^{78}Ni with the VS-IMSRG-based interaction”, PCM2025, Aizu Univesity, Aizu-wakamatsu, Japan, March 3-7, 2025.

17 **Contributed talk** on “Exploring the Structure of N = 50 Nuclei through the Monte Carlo Shell Model”, DAE-BRNS Symposium on Nuclear Physics, NIT Jalandhar, December7-12, 2025.

18 **Contributed talk** on “Probing the shape evolution and shell structures in neutron-rich N=50 nuclei”, Microscopic Description of Nuclear Collective Phenomena via Large-Scale Density Functional Calculations, Hokkaido University, Sapporo, Japan, January 5 - 9, 2026.

Talks Delivered/ Poster Presentations (continued)

19 **Contributed talk** on "Probing the shape evolution and shell structures in neutron-rich N=50 nuclei", Symposium of Basic Science Research Programs on the Fugaku supercomputer, Tokyo, Japan, January 13-15, 2026.

20 **Invited talk** on "Nuclear Shell Structure and Its Evolution in Neutron-rich Nuclei", Nuclear Theory in the Supercomputing Era NTSE–2026, IIT Roorkee, India, March 16-20, 2026.

21 **Contributed talk** on "Exploring the Nuclear Shell Model with Quantum Computing", Third RIKEN Quantum internal meeting", RIKEN, Wako, Japan, March 30, 2026.

22 **Poster present** on "Probing the shape evolution and shell structures in neutron-rich N=50 nuclei", in the International confrence "Advances in Radioactive Isotope Science (ARIS2026)", Fukushima, Japan, June 6, 2026.

Awards and Achievements

2024 **Joint Institute for Computational Fundamental Science (JICFuS)** published a magazine about me in Japan society.

2020 **Recipient of International travel grant** from Science and Engineering Research Board (SERB), India, to present work in International Conference on Neutrino and Nuclear Physics 2020 (CNNP2020) held at iThemba LABS, Cape Town (South Africa), 24 - 28 Feb, 2020.

2016 **Recipient of Senior Research Fellowship** from Ministry of Human Resource and Development (MHRD), India, December, 2018- December, 2021.

Recipient of Junior Research Fellowship from Ministry of Human Resource and Development (MHRD), India, December, 2016- December, 2018.

Qualified Graduate Aptitude Test in Engineering (GATE) in Physics.

2013 **Merit and Scholarship Awarded** by the "R. R. Morarka Charitable Trust" for the got Merit in B.Sc. final year during 2012-13.

Computational Skills

Operating System	Linux and Windows.
Coding languages	Fortran (77, 90 and 95), C++, Python.
Parallel Programming	OpenMP & MPI.
Nuclear Shell-Model Code	Kshell, NuShellX, Bigstick, Oxbash and Antoine.
Software	Matlab, MS Office, Libre Office, LaTeX, and Beamer.
Graphics	Origion, XmGrace, GNU Plot, Matplotlib, Plotly, and Pgfplots.
Misc.	Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

Computational Skills (continued)

Developed Code

- (I). During Ph.D., I have developed my own computational codes in Fortran90 for the calculation of nuclear weak-transition matrix elements associated with allowed and forbidden β-decays. The code enables the evaluation of partial half-lives, electron spectral shapes, and shape factors, including next-to-leading-order corrections that are particularly important for forbidden β-decay transitions. This computational framework has been successfully applied to systematic studies of nuclear weak processes across different regions of the nuclear chart and has contributed to several peer-reviewed publications.
(II). During my postdoctoral career, I developed an efficient nuclear shell-model code in Python, named **PyShell**, which is designed to run on both CPU and GPU architectures. The primary motivation behind its development was to provide a flexible platform for implementing advanced *Artificial Intelligence (AI)* and *Quantum Computing (QC)* techniques in nuclear shell-model calculations.

Advanced Pre-PhD Courses

- **Advanced Classical Physics**
Passed with 4/4 Credits (January - May , 2017)
held at Indian Institute of Technology Roorkee.

- **Mathematical And Computational Techniques**
Passed with 4/4 Credits (July - November , 2017)
held at Indian Institute of Technology Roorkee.

- **Advanced Quantum Physics**
Passed with 4/4 Credits (January - May , 2017)
held at Indian Institute of Technology Roorkee.

- **Computational Nuclear Physics**
Passed with 4/4 Credits (January - May , 2017)
held at Indian Institute of Technology Roorkee.

Teaching Experience

January 2020 – April 2020

- **Teaching Assistant,** in experimental lab of Physics (Optics, Electronics and Modern Physics experiments) for first year engineering undergraduates at Indian Institute of Technology Roorkee.

July 2019 – November 2019

- **Teaching Assistant,** in MATLAB (subject code PH-701) coding class (computer lab) for first year M. Tech. students at Indian Institute of Technology Roorkee.

January 2019 – May 2019

- **Teaching Assistant,** in theory class of Mechanics and Relativity (subject code PHN-104) for first year engineering undergraduates at Indian Institute of Technology Roorkee.

Teaching Experience (continued)

July 2018 – November 2018

- **Teaching Assistant,** in MATLAB (subject code PH-701) coding class (computer lab) for first year M. Tech. students at Indian Institute of Technology Roorkee.
- **Teaching Assistant,** in experimental lab of Nuclear Physics (subject code PH-607) for M.Sc. Second Year & Int. M.Sc. Fifth year students, at Indian Institute of Technology Roorkee.

January 2018 – May 2018

- **Teaching Assistant,** in theory class of Mechanics and Relativity (subject code PHN-104) for first year engineering undergraduates at Indian Institute of Technology Roorkee.
- **Teaching Assistant,** in experimental lab of Physics (Optics, Electronics and Modern Physics experiments) for first year engineering undergraduates at Indian Institute of Technology Roorkee.

July 2017 – November 2017

- **Teaching Assistant,** in theory class of Electromagnetic theory (subject code PHN-005) for first year engineering undergraduates at Indian Institute of Technology Roorkee.
- **Teaching Assistant,** in experimental lab of Physics (Optics, Electronics and Modern Physics experiments) for first year engineering undergraduates at Indian Institute of Technology Roorkee.

Other Experiences and Activities

2017-2018

- Served as the **Culture Secretary of the Physics Association** in the Department of Physics, Indian Institute of Technology, Roorkee.

Personal Details

- Name: Anil Kumar
- Father's name: Natthu Ram
- Mother's name: Vimla Devi
- Date of birth: August 16, 1993
- Place of birth: Jhunjhunu, Rajasthan, India
- Current Address: Ibaraki-ken, Tsukuba-shi, Kasuga 4-14-22, Villa Shoei A101, Japan.
- Permanent Address: Khicharon Ka Bass, Jhunjhunu, Rajasthan, India
- Nationality: Indian
- Gender: Male
- Marital Status: Unmarried
- Languages Known: English and Hindi (Native)

References

Prof. Praveen C. Srivastava
Ph.D. Thesis Supervisor
Department of Physics,
IIT Roorkee, India
praveen.srivastava@ph.iitr.ac.in

Prof. Noritaka Shimizu
(Postdoc Supervisor)
Center for Computational Sciences
University of Tsukuba, Japan
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Prof. Yutaka Utsuno
(Collaborator)
Japan Atomic Energy Agency
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Prof. Takayuki Miyagi
(Collaborator)
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miyagi@nucl.ph.tsukuba.ac.jp

Prof. Vikas Kumar
(Collaborator)
Department of Physics
Banaras Hindu University, India
vikasphysics@bhu.ac.in

Declaration: I hereby declare that the particulars furnished herein by me are true to the best of my knowledge and belief.

Place: Jhunjhunu
Date:

(Dr. Anil Kumar)